

## SECTION 23 05 19 - METERS AND GAUGES FOR HVAC PIPING

### PART 1 - GENERAL

#### 1.01 REFERENCE STANDARDS

- A. ASME B40.100 - Pressure Gauges and Gauge Attachments 2013.
- B. ASME MFC-3M - Measurement of Fluid Flow in Pipes Using Orifice, Nozzle and Venturi 2004 (Reaffirmed 2017).
- C. ASTM E1 - Standard Specification for ASTM Liquid-in-Glass Thermometers 2014.
- D. ASTM E77 - Standard Test Method for Inspection and Verification of Thermometers 2014, with Editorial Revision (2017).
- E. AWWA C700 - Cold-Water Meters -- Displacement Type, Metal Alloy Main Case 2015.
- F. AWWA M6 - Water Meters -- Selection, Installation, Testing, and Maintenance 2012, with Addendum (2018).
- G. UL 393 - Indicating Pressure Gauges for Fire-Protection Service Current Edition, Including All Revisions.

#### 1.02 SUBMITTALS

- A. Product Data: Provide list that indicates use, operating range, total range and location for manufactured components.

#### 1.03 GENERAL INFORMATION AND DESIGN GUIDELINES

- A. Requirements: Refer to CSU Building Metering Guide. Website:  
[http://www.calstate.edu/cpdc/ae/gsf/documents/CSU\\_Metering\\_Guide.pdf](http://www.calstate.edu/cpdc/ae/gsf/documents/CSU_Metering_Guide.pdf)
- B. Intent: Metering to meet the performance-based LEED NC/EB Energy & Atmosphere Credit – Building-level energy metering standards for ongoing accountability and optimization of building energy and water consumption performance over time.
- C. Refer to and coordinate with Section 25 00 00 – Integrated Automation.

- D. Metering Equipment: Allow Campus the ability to identify problems and achieve improved system performance. Select meters for future connection to a remote-read automated metering network.
- E. BTU meter shall be installed at each building and for each system (heating, cooling and domestic hot water). Onicon System 10, F-3200 mag BTU meter, or approved equal.
- F. Flow Meters:
  - 1. Function: Non-invasive flow measurement with a transmitter with 2-line backlit display with 4-button keypad. Connect to Campus Energy Management System (EMS) by Siemens.
  - 2. Execution: Locate on straight run of pipe with no flow interruptions such as valves or direction change. Installed location shall typically have straight uninterrupted pipe for 10 diameters upstream and 5 diameters downstream. Consult manufacturer's installation instructions for specific installation requirements and other piping configurations.
- G. Keys for Cabinets and Padlocks:
  - 1. Cabinets and Equipment: Provide 2 keys per panel. Coordinate with Section 08 06 05 – Key Schedule.
  - 2. Padlocks: Coordinate with Section 08 06 05 – Key Schedule.
  - 3. Closeout Submittal: Provide panel keys separated and labeled. Provide location, room number, quantity, manufacturer name and model numbers of keys, and coordinate closeout submittal with Section 08 06 05 – Key Schedule.

## PART 2 - PRODUCTS

### 2.01 POSITIVE DISPLACEMENT METERS (LIQUID)

- A. AWWA C700, positive displacement disc type suitable for fluid with metal alloy main case and cast-iron frost-proof, breakaway bottom cap, hermetically sealed register, remote reading.
- B. Meter: Brass body turbine meter with magnetic drive register.
  - 1. Service: Cold water, 122 degrees F (50 degrees C).
  - 2. Service: Hot water, 200 degrees F (93 degrees C).

3. Accuracy: 1-1/2 percent.

## 2.02 HEAT CONSUMPTION METERS

- A. Meter: Brass body turbine meter with magnetic drive register, platinum temperature sensors.
  1. Maximum Service Temperature: 200 degrees F (93 degrees C).
  2. Accuracy: 1-1/2 percent.

## 2.03 LIQUID FLOW METERS

- A. Calibrated ASME MFC-3M Venturi orifice plate and flanges with valved taps, chart for conversion of differential pressure readings to flow rate, with pressure gauge in case.
- B. Annular element flow stations with meter set.
  1. Measuring Station: Type 316 stainless steel pitot type flow element inserted through welded threaded couplet, with safety shut-off valves and quick coupling connections, and permanent metal tag indicating design flow rate, reading for design flow rate, metered fluid, line size, station or location number.
    - a. Pressure rating: 275 psi (1896 kPa).
    - b. Maximum temperature: 400 degrees F (204 degrees C).
    - c. Accuracy: Plus 0.55 percent to minus 2.30 percent.

## 2.04 PRESSURE GAUGES

- A. Pressure Gauges: ASME B40.100, UL 393 drawn steel case, phosphor bronze bourdon tube, rotary brass movement, brass socket, with front recalibration adjustment, black scale on white background.
  1. Mid-Scale Accuracy: One percent.
  2. Scale: Psi and KPa.

## 2.05 PRESSURE GAUGE TAPPINGS

- A. Gauge Cock: Tee or lever handle, brass for maximum 150 psi (1034 kPa).
- B. Needle Valve: Brass, 1/4 inch (6 mm) NPT for minimum 150 psi (1034 kPa).

- C. Pulsation Damper: Pressure snubber, brass with 1/4 inch (6 mm) connections.
- D. Syphon: Steel, Schedule 40, 1/4 inch (6 mm) angle or straight pattern.

## 2.06 STEM TYPE THERMOMETERS

- A. Thermometers - Fixed Mounting: Red- or blue-appearing non-toxic liquid in glass; ASTM E1; lens front tube, cast aluminum case with enamel finish.
  - 1. Accuracy: 2 percent, per ASTM E77.
  - 2. Calibration: Degrees F.
- B. Thermometers - Adjustable Angle: Red- or blue-appearing non-toxic liquid in glass; ASTM E1; lens front tube, cast aluminum case with enamel finish, cast aluminum adjustable joint with positive locking device; adjustable 360 degrees in horizontal plane, 180 degrees in vertical plane.
  - 1. Accuracy: 2 percent, per ASTM E77.
  - 2. Calibration: Degrees F.

## 2.07 DIAL THERMOMETERS

- A. Thermometers - Fixed Mounting: Dial type bimetallic actuated; ASTM E1; stainless steel case, silicone fluid damping, white with black markings and black pointer, hermetically sealed lens, stainless steel stem.
  - 4. Accuracy: 1 percent.
  - 5. Calibration: Degrees F.
- B. Thermometer: ASTM E1, stainless steel case, adjustable angle with front recalibration, bimetallic helix actuated with silicone fluid damping, white with black markings and black pointer hermetically sealed lens, stainless steel stem.
  - 6. Accuracy: 1 percent.
  - 7. Calibration: Degrees F.

## 2.08 STATIC PRESSURE GAUGES

- A. 3-1/2 inch (90 mm) diameter dial in metal case, diaphragm actuated, black figures on white background, front recalibration adjustment, 2 percent of full scale accuracy.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. Install positive displacement meters with isolating valves on inlet and outlet to AWWA M6. Provide full line size valved bypass with globe valve for liquid service meters.
- B. Provide instruments with scale ranges selected according to service with largest appropriate scale.

END OF SECTION 23 05 19